# Akoizo Original Practice - Form 1 - Answer key

### Division C | Experimental Design | 50-minute written practice

Original Akoizo curriculum practice. This is not a historical tournament paper. Written build, CAD, lab and communication exercises supplement real practical preparation.

### ANSWER KEY - Review only after submitting.

**1. (2 points) Which is most operational?**

Unit: The experiment and the prompt packet.
Worked-case data: A prompt supplies a ramp, object, ruler, and timer. Turn “study motion” into a testable question.

A. Time in seconds between two fixed marks

B. How good the motion is

C. Whether the object seems fast

D. Interesting behavior

**Correct: A**

**2. (2 points) A fixed +0.5 s offset is…**

A. A new independent variable always

B. No error

C. Systematic bias

D. Only random scatter

**Correct: C**

**3. (2 points) Which is a direct qualitative observation?**

Unit: Materials, reproducible procedures, and qualitative data.
Worked-case data: A trial has no timer reading because the observer missed the endpoint. What should the table and procedure record?

A. Friction definitely increased

B. The hypothesis is true

C. The cause was gravity alone

D. The object wobbled

**Correct: D**

**4. (2 points) A missing trial should be fabricated to complete the table?**

Unit: Practice, timing, teamwork, and final review.
Worked-case data: A team finishes data collection with little time left and no graph or analysis. Suggest a targeted practice change.

A. Only if software predicts it

B. No

C. Yes

D. Only if the hypothesis is clear

**Correct: B**

**5. (2 points) A source of error explanation should include…**

Unit: Claims, evidence, and reasoning for trends and outliers.
Worked-case data: At levels 1,2,3, means are 12,16,20 units, with SD 2 at each level. Write a bounded trend CER.

A. Only a list of tools

B. Only an apology

C. How it changes the response

D. Only the word “human”

**Correct: C**

**6. (4 points) Write a directional ramp hypothesis.**

Unit: Problem, hypothesis, variables, and experimentation.
Worked-case data: All low-level trials occur early and all high-level trials occur late while the instrument drifts upward.

answer: Increasing release height will reduce defined transit time under comparable conditions because available potential energy increases. (4 points).

**7. (4 points) Report 2.0 + 3.00 cm.**

answer: 5.0 cm using the limiting tenths-place precision. (4 points).

**8. (4 points) Distinguish SD and standard error.**

Unit: Quantitative data, graphs, and statistics.
Worked-case data: For response values 8, 10, 12, calculate mean, range, and sample SD.

answer: SD describes individual-value spread; standard error estimates mean precision under valid independence assumptions. (4 points).

**9. (4 points) Why keep raw data?**

Unit: The experiment and the prompt packet.
Worked-case data: A prompt supplies a ramp, object, ruler, and timer. Turn “study motion” into a testable question.

answer: They allow auditing, recalculation, plotting, and transparent handling of failed trials. (4 points).

**10. (4 points) Recommend a targeted improvement for reaction-time variation.**

Unit: Errors, conclusions, and recommendations.
Worked-case data: Every timer reading is 0.5 s too high. Would ten more repeats eliminate the bias?

answer: Use a defined automated endpoint or more consistent timing method, then verify its calibration and applicability. (4 points).

**11. (6 points) Solve this case and explain your reasoning: A trial has no timer reading because the observer missed the endpoint. What should the table and procedure record?**

reasoning: Mark the value missing, not zero.
Record the missed endpoint as the reason and retain the trial identifier.
Repeat under a predefined rule if feasible, documenting the repeat and its relation to the failed trial. (3 points).

conclusion: Transparent missing-data handling preserves the evidence trail. (3 points).

**12. (4 points) Give a targeted follow-up for excessive setup time.**

Unit: Practice, timing, teamwork, and final review.
Worked-case data: A team finishes data collection with little time left and no graph or analysis. Suggest a targeted practice change.

answer: Use a simpler apparatus, pilot feasibility early, coordinate independent tasks, and reserve analysis time in a new practice run. (4 points).

**13. (6 points) Solve this case and explain your reasoning: At levels 1,2,3, means are 12,16,20 units, with SD 2 at each level. Write a bounded trend CER.**

reasoning: Claim: mean response increases across the tested range.
Evidence: means rise by 4 units per level and by 8 overall, with SD about 2.
Reasoning: a proposed increasing mechanism is consistent, but design validity and sample size limit certainty and extrapolation. (3 points).

conclusion: A complete CER quantifies support and states what it does not establish. (3 points).

**14. (2 points) Define “Control condition” in the context of Experimental Design.**

answer: Reference treatment for comparison. (2 points).

**15. (2 points) Define “Standard of comparison” in the context of Experimental Design.**

answer: Defined reference condition for evaluating treatments. (2 points).

**16. (2 points) Define “Sample standard deviation” in the context of Experimental Design.**

answer: Spread estimate using squared deviations and n - 1. (2 points).

**17. (2 points) Define “Operational definition” in the context of Experimental Design.**

answer: Specific rule for how a variable is set or measured. (2 points).

**18. (2 points) Define “Validity” in the context of Experimental Design.**

answer: How well evidence supports the intended inference. (2 points).